

Week 5: CI/CD Automation with Azure DevOps

Objective

Automate smart home energy report generation using an Azure DevOps pipeline that fetches new data, cleans it, summarizes energy usage, and alerts when device usage exceeds a threshold.

Capstone Tasks

- Automate the energy report generation weekly
- Build a pipeline to fetch new data, clean, and summarize
- Alert or log if usage crosses a threshold (e.g., >10 kWh per device per day)

Project Files

energy_report.py

```
import pandas as pd
```

```
# Load energy data
```

```
df = pd.read_csv('energy_usage.csv')
```

```
# Summarize energy per device
```

```
device_summary = df.groupby('device_id')['energy_kwh'].sum().reset_index()
```

```
# Save report
```

```
device_summary.to_csv('weekly_energy_report.csv', index=False)
```

```
print('Weekly energy report generated successfully!')
```

```
# Alert for high usage
```

```
threshold = 10
```

```
high_usage = device_summary[device_summary['energy_kwh'] > threshold]
```

```
if not high_usage.empty:

    print('ALERT: Some devices exceeded 10 kWh usage!')

    print(high_usage)

else:

    print('All devices are within the energy limit.')
```

energy_usage.csv

```
device_id,room_id,timestamp,energy_kwh

1,101,2026-05-13 10:00:00,2.5

1,101,2026-05-14 11:00:00,3.0

2,102,2026-05-13 10:10:00,5.2

2,102,2026-05-14 11:10:00,4.8

3,103,2026-05-13 10:20:00,1.1

3,103,2026-05-14 11:20:00,2.0
```

Azure DevOps YAML Pipeline

Create a file named azure-pipelines.yml

trigger:

- main

schedules:

- cron: "0 0 * * 0"

displayName: Weekly Energy Report

branches:

include:

- main

always: true

pool:

vmImage: 'ubuntu-latest'

steps:

- task: UsePythonVersion@0

inputs:

versionSpec: '3.x'

- script: |

pip install pandas

python energy_report.py

displayName: 'Run Smart Energy Analysis'

- task: PublishBuildArtifacts@1

inputs:

pathToPublish: 'weekly_energy_report.csv'

artifactName: 'EnergyReport'

Pipeline Workflow

1. Trigger from the main branch.
2. Schedule weekly execution.
3. Install Python dependencies.
4. Run the energy analysis script.
5. Generate weekly_energy_report.csv.
6. Print alert if any device exceeds 10 kWh.
7. Publish the report as a pipeline artifact.

Expected Output

Console Log

Weekly energy report generated successfully!

ALERT: Some devices exceeded 10 kWh usage!

Output File

weekly_energy_report.csv with device-wise energy totals.

Deliverables

- Azure DevOps pipeline YAML
- Execution log
- Weekly energy report output